Hematology Image Recognition + High-Yield Genetics

Hematology Image Recognition

Acute Promyelocytic Leukemia(APL)

Image clue:

- Promyelocytes with abundant granules
- Multiple Auer rods("Faggot cells")

Genetics:

- t(15;17)
- PML-RARA

Buzzword:

- DIC
- All-trans retinoic acid(ATRA)

Acute Myeloid Leukemia(AML)

Image clue:

- Myeloblasts with Auer rods

Genetics:

- t(8;21)
- inv(16)

- t(15;17)(APL subtype)

Buzzword:

- Auer rods = AML until proven otherwise

Chronic Myeloid Leukemia(CML)

Image clue:

- Granulocytes at all stages of maturation
- Myelocytes,metamyelocytes,neutrophils together

Genetics:

- t(9;22)
- BCR-ABL
- Philadelphia chromosome

Chronic Lymphocytic Leukemia(CLL)

Image clue:

- Smudge cells

Association:

- Elderly patient

- Absolute lymphocytosis

Hairy Cell Leukemia

Image clue:

- Lymphocytes with hair-like cytoplasmic projections

Genetics:

- BRAF mutation

Hodgkin Lymphoma

Image clue:

- Reed-Sternberg cells
- Owl-eye appearance

Burkitt Lymphoma

Image clue:

- Starry-sky appearance

Genetics:

- t(8;14)

- MYC

Follicular Lymphoma

Image clue:

- Nodular/follicular growth pattern

Genetics:

- t(14;18)
- BCL2

Mantle Cell Lymphoma

Genetics:

- t(11;14)
- Cyclin D1(CCND1)

One-Line Revision Table

Finding	Diagnosis
Auer rod	AML
Faggot cell	APL
Smudge cell	CLL
Hairy projections	Hairy cell leukemia
Reed-Sternberg cell	Hodgkin lymphoma
Starry-sky appearance	Burkitt lymphoma
Philadelphia chromosome	CML

High-Yield Genetics for NEET PG / INI-CET

Tier 1: Must Know

Disease	Gene/Mutation
CML	BCR-ABL,t(9;22)
APL	PML-RARA,t(15;17)
Burkitt lymphoma	MYC,t(8;14)
Follicular lymphoma	BCL2,t(14;18)
Mantle cell lymphoma	Cyclin D1(CCND1),t(11;14)
Hairy cell leukemia	BRAF
GIST	KIT(CD117)
Retinoblastoma	RB1
Li-Fraumeni syndrome	TP53
Familial adenomatous polyposis	APC
MEN 2	RET
Neurofibromatosis Type 1	NF1
von Hippel-Lindau syndrome	VHL
Tuberous sclerosis	TSC1, TSC2

Tier 2: Very Common

Disease	Gene
CLL	del 13q
AML	FLT3, NPM1
Polycythemia vera	JAK2
Essential thrombocythemia	JAK2
Primary myelofibrosis	JAK2
Diffuse large B-cell lymphoma	BCL6
Medullary thyroid carcinoma	RET
Papillary thyroid carcinoma	RET/PTC,BRAF
Pheochromocytoma	RET,VHL,NF1
Clear cell RCC	VHL
Wilms tumor	WT1
Ewing sarcoma	t(11;22),EWSR1-FLI1
Synovial sarcoma	t(X;18)
Alveolar rhabdomyosarcoma	t(2;13)

Tier 3: Common Solid Tumors

Disease	Gene
Colorectal carcinoma	APC → KRAS → TP53
Lynch syndrome	MLH1,MSH2,MSH6,PMS2
Pancreatic adenocarcinoma	KRAS
Melanoma	BRAF
Lung adenocarcinoma	EGFR, ALK
Breast carcinoma	BRCA1,BRCA2,HER2
Ovarian carcinoma	BRCA1, BRCA2
Endometrial carcinoma	PTEN
Basal cell carcinoma	PTCH
Medulloblastoma	SHH/PTCH pathway

The Chromosome 14 Rule

Chromosome 14 contains the Immunoglobulin Heavy Chain(IgH)locus.

Translocation	Disease	Gene
t(8;14)	Burkitt lymphoma	MYC
t(14;18)	Follicular lymphoma	BCL2
t(11;14)	Mantle cell lymphoma	Cyclin D1

Tumor Suppressor Syndromes

Syndrome	Gene
Retinoblastoma	RB1
Li-Fraumeni	TP53
FAP	APC
VHL	VHL
NF1	NF1
Tuberous sclerosis	TSC1, TSC2
Lynch syndrome	DNA mismatch repair genes

JAK2 Group

Disease	Mutation
Polycythemia vera	JAK2
Essential thrombocythemia	JAK2
Primary myelofibrosis	JAK2

Reverse Memorization Table(Most Useful)

Gene	Disease
BCR-ABL	CML
PML-RARA	APL
MYC	Burkitt lymphoma
BCL2	Follicular lymphoma
Cyclin D1	Mantle cell lymphoma
JAK2	PV,ET,PMF
RET	MEN2, Medullary thyroid carcinoma
APC	FAP
TP53	Li-Fraumeni syndrome
RB1	Retinoblastoma
BRAF	Hairy cell leukemia, Melanoma
KIT(CD117)	GIST
VHL	Clear cell RCC
WT1	Wilms tumor
EGFR	Lung adenocarcinoma
HER2	Breast carcinoma
BRCA1/2	Breast and ovarian carcinoma